

Schoolyard Plant Detectives

An observation-based investigation that helps learners compare local plants, identify patterns, and support a claim with evidence.

GRADES 3 to 5	TIME 45 to 60 minutes	MATERIALS Paper, pencil, string	SETTING Schoolyard or window view
-------------------------	---------------------------------	---	---

Learning goals

By the end of the investigation, learners should be able to:

- observe plant structures using safe, non-destructive methods;
- record similarities and differences in a simple table;
- make a claim about how a plant is suited to its location and support it with observations.

Safety first

Set clear boundaries. Learners should not taste plants, touch unknown sap, climb, or remove living parts. Observe from a safe distance when needed.

Before class

Choose two or three plants that learners can observe safely. Include plants from different spots if possible, such as sun and shade, dry and damp ground, or open and sheltered areas.

If going outside is not possible, use plants visible from a window or bring safe fallen leaves. Learners can also compare clear teacher-made sketches.

Observe, compare, and explain

STEP 1 Notice	STEP 2 Record	STEP 3 Compare	STEP 4 Claim
-------------------------	-------------------------	--------------------------	------------------------

1. Notice without naming - 8 minutes

Ask learners to silently observe one plant. They list colors, shapes, textures, size, and where the plant is growing. Avoid giving the plant name first so the focus stays on evidence.

2. Record two plants - 15 minutes

Pairs record leaf shape, stem type, approximate height, nearby light, and soil condition. They may add one question they cannot answer through observation alone.

3. Compare patterns - 10 minutes

Learners circle one similarity and underline one difference. Ask: Which observation can another person verify? Which statement is only a guess?

4. Make an evidence-based claim - 12 minutes

Use the frame: I think this plant is suited to ____ because I observed _____. A useful follow-up is: What other evidence would make your claim stronger?

Multigrade move

Grade 3: identify visible parts. Grade 4: compare structures and locations. Grade 5: write a claim with two pieces of evidence and note one limitation.

FIELD NOTES

Plant detective record

Name: _____ Grade: _____ Date: _____

Observation	Plant A	Plant B
Leaf shape and texture		
Stem or trunk		
Light and ground		
One other detail		

My claim

I think _____ is suited to _____ because I observed _____.

A question for our next investigation

What do you still want to find out?
